Intelligent Consensus Asymptotic Tracking Control for Nonlinear Multiagent Systems Under Denial-of-Service Attacks

Yuqiang Jiang, Ben Niu, Tao Zhao, Xudong Zhao, Xinjun Wang, and Huanqing Wang

***Abstract*— This work considers the adaptive output feedback consensus asymptotic tracking control problem for full-state-constrained nonlinear multiagent systems (MASs) under the denial-of-service (DoS) attacks and the actuator faults. Since MASs are regardered as a network, where each agent is considered as a network node, the DoS attacks that cut network communication links between agents are studied in this paper. Moreover, to compensate the unknown actuator fault signal, the Nussbaum function method is introduced into the controller design process for each agent. Due to the researched nonlinear MASs contain unknown nonlinearities and the system states of each agent are unmeasurable, an estimator based on fuzzy logic systems (FLSs) is constructed for each agent. Further, the secure control strategy based on the barrier Lyapunov functions (BLFs) is proposed for the researched nonlinear MASs, which guarantees the consensus tracking errors asymptotically converge to zero, all closed-loop signals remain bounded and the full-state constraint requirements are not broken. In the illustrative example, the proposed secure control strategy is applied to the MASs composed of multiple forced damped pendulums (FDPs).**

***Note to Practitioners*— In the framework of network communication, how to eliminate the impact of cyber attacks on nonlinear MASs that rely on communication links to transmit information is critical. Furthermore, due to practical systems are inevitably subject to multiple physical and environmental constraints, the problem of dealing with system constraints can not be ignored. Therefore, a secure control algorithm based on the BLFs is proposed to eliminate the effects of the DoS attacks on the researched nonlinear MASs and ensure that the full-state constraints are not broken. To realize the output feedback control and approximate the unknown nonlinearities of the studied system, the estimators combining with the intelligent approximation techniques based on FLSs are developed. The Nussbaum function is introduced into controller to solve the actuator faults problems for each agent.**

***Index Terms*— Nonlinear multiagent systems, consensus asymptotic tracking, DoS attacks, full-state constraints, state estimator.**

I. Introduction

THE burgeoning application potential of MASs has prompted a surge in interest regarding the consensus control problem for MASs [1], [2], [3], [4]. Given that numerous real-world systems exhibit intricate nonlinear dynamics, the focus of research has recently shifted from linear MASs to nonlinear MASs [5], [6], [7], [8]. According to the different communication relationships between agents, consensus control can be divided into two categories: leaderless consensus control [9] and [10] and leader-following consensus control [11], [12], [13]. It is shown in [1] and [3] that the leader-following consensus control is more practical than the leaderless consensus control. Due to the presence of the communication topology and the limited sensing capability of sensors, all agents under the leader-following control framework should be completely distributed, which means each agent does not require the information of all agents, relies only on the partial information from its neighbors. It is worth noting that when the neighbor's information is maliciously attacked, the consensus performance of nonlinear MASs will be threatened. Therefore, how do we maintain the ideal consensus performance of the attacked nonlinear MASs is crucial.

As nonlinear MASs function concurrently within both the physical and network layers, they present enormous challenges for practical applications when facing attacks from both layers. Within the vulnerable network layer, transmitted data becomes susceptible to interference and tampering by malicious actors, who can block or alter the information. Therefore, the network attacks are classified as DoS attacks [15] and deception attacks [16]. It is important to point out the DoS attacks on MASs means that the attackers block the followers from receiving the information sent from their neighbors. A number of control strategies have been proposed, such as a switching-type adaptive state estimator was constructed for the nonlinear uncertain interconnected systems to overcome the difficulty caused by DoS attacks in [17]. Reference [18] proposed an adaptive event-triggered secure control method

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for the stochastic nonlinear high-order MASs subject to DoS attacks. However, the asymptotic stability of systems under DoS attacks still deserves further study. Within the physical layer, equipment disrepair and faulty operation can lead to a decrease in the effectiveness of system controllers. This decrease in effectiveness has the potential to result in consensus degradation or, in more severe cases, even lead to serious accidents. It is widely recognized that fault-tolerant methods have been proposed to effectively address issues stemming from actuator faults. For example, a fault-tolerant controller was developed for the MASs under the actuator bias faults in [14]. A distributed fault-tolerant solution was proposed for the networks linear MASs under the multiple heterogeneous actuator faults in [19]. Moreover, in the most existing results [14], [17], [18], [19], the proposed control strategies only focused on the attacks from a single layer (network layer or physical layer). This motivates us to explore the anti-attack research on multiple layers.

It is worth noting that the communication between each agent and its neighbor depends on the channel of the network. Thus, the channel resources are crucial to MASs. To save the channel resources effectively, the event-triggered control strategy is gradually developed to some related works on the MASs [20], [21], [22], [23], which determine the data transmission and control update via the event-triggered conditions. On the other hand, many actual systems often face constraints imposed by their operational environments or the necessity for safe operation, such as systems with nonholonomic wheel [24] and vehicle systems actuated by four wheel [25]. How to ensure that the state constraint requirements are not destroyed is crucial, and some works have been obtained [26], [27], [28]. For example, [29] introduced the one-to-one nonlinear mapping method, where the constrained system in strict-feedback form was changed as a unconstrained pure feedback system. Reference [30] proposed the novel BLFs, which not only ensure that the requirements of the state constraints are not destroyed, but also maintain that the tracking performance is ideal.

According to the discussions above, this work solves the estimator-based adaptive consensus asymptotic tracking control problem for the researched nonlinear MASs. The main contributions are summarized in the following points. ***(i)*** In the presence of concurrent DoS attacks and actuator faults, and when only the output information of each agent is accessible, a novel distributed consensus asymptotic tracking control strategy based on the fuzzy state estimator is developed for the nonlinear MASs. ***(ii)*** Different from the existing fault-tolerant strategies [14] and [19], our approach distinctly incorporates the Nussbaum function into the controller design process, thereby effectively compensating for performance degradation under both normal and fault conditions. Furthermore, one adaptive law is designed to eliminate the effect of the lock-in-place fault in this paper. ***(iii)*** Unlike the existing works [31], [32], [33], [34] which utilize time-triggered approaches, our approach integrates a relative threshold event-triggered strategy into the anti-attack controller for each agent. This not only efficiently preserves communication resources but also ensures that consensus tracking errors asymptotically converge to zero, all closed-loop signals remain within bounds, and the state constraint requirements of each agent are not violated.

II. Preliminaries and Problem Formulation

A. Graph Theory

The directed graph $G = (V, E)$ is defined to represent the topology relationship between agents, where $V = \{1, \ldots, N\}$ is a node set of agents and $E \in V \times V$ is a edge set. When any point is chosen as the root node, the graph G contains a directed spanning tree. The edge $(i, j) \in E$ represents that the jth agent can pass information to its neighbor jth agent. The direct topology graph $G = (V, E)$ is described as the matrix $A = [b_{i,j}] \in R^{N\times N}$, where the matrix A is also called the adjacency matrix of G. If the edge (i, j) is connected, the element $b_{i,j} = 1$; otherwise, the element $b_{i,j} = 0$. The in-degree matrix $D = diag\{\Xi_1, \ldots, \Xi_N\}$, where $\Xi_i = \sum_{j=1}^{N} b_{i,j}$. Futher, the Laplacian matrix $L = D - A$, which has a characteristic that each row sum is zero. Moreover, the leader-following diagonal matrix is defined as $B = diag\{b_i\} \in R^{N\times N}$ where b_i represents the weight of the ith follower and leader, $i = 1, \ldots, N$.

B. System Descriptions and Assumptions

Consider the following dynamics of the ith follower:

$$\begin{cases} \dot{x}_{i,m} = x_{i,m+1} + f(\bar{x}_{i,m}), \\ \dot{x}_{i,l} = u_i^f + f(\bar{x}_i), \\ y_i = x_{i,1}, \end{cases} \tag{1}$$

where $\bar{x}_{i,m} = [x_{i,1}, \ldots, x_{i,m}]^T$, $\bar{x}_i = [x_{i,1}, \ldots, x_{i,l}]^T (i = 1, \ldots, N, m = 1, \ldots, l-1)$ are the state vector of the ith follower. u_i^f and y_i are the control input and the output of the ith follower, respectively. $f(\bar{x}_{i,m})$ and $f(\bar{x}_i)$ are both unknown smooth nonlinear functions. In this paper, all the states are constrained in the compact sets, i.e., $|x_{i,m}| \le k_{i,c_m}$, where k_{i,c_m} is the positive constant.

The purpose of this paper is to design a distributed anti-attack controller for the nonlinear MASs (1) under the DoS attacks, such that:

(i) Each follower's output asymptotically converges to the leader's trajectory.

(ii) All signals in the closed-loop system are bounded.

To ensure the integrity and conciseness of the control design process, the following assumptions and lemmas are given.

Assumption 1: For any $k_{i,c_1} > 0$, given a constant $y_0 > 0$, such that the desired trajectory y_d satisfies $|y_d| \le y_0 < k_{i,c_1}$ and its time derivatives are continuous up to the l-th order.

Assumption 2: Given a set of constants $L_{i,j}$, $j=1, \cdots, l$, for $\forall \upsilon_{i,1}, \upsilon_{i,2} \in R^i$ such that the following inequality holds:

$$\left| f_{i,j}(\upsilon_{i,1}) - f_{i,j}(\upsilon_{i,2}) \right| \le L_{i,j} \left\| \upsilon_{i,1} - \upsilon_{i,2} \right\|, \tag{2}$$

where $\|\cdot\|$ denotes the 2-norm.

Lemma 1 [35]: Given a continuous function $F(\cdot)$ and assuming its integration $\lim_{t\to\infty} \int_0^t F(\psi) d\psi$ exists and is finite. Then, $\lim_{t\to\infty} F(t) = 0$ is valid.

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Lemma 2 *[36]:* The smooth functions $V(t) \geq 0$, $\varpi(t)$ and $N(\varpi)$ are defined on $[0, t_f]$, where $N(\varpi)$ is the Nussbaum-type function. If the following inequality holds:

$$V(t) \leq c_1 + e^{-c_2 t} \int_0^t [g(\tau)N(\varpi)\dot{\varpi} + 1]e^{c_2 t} d\tau, \tag{3}$$

where c_1, c_2 are the positive constants and $g(\tau)$ is a non-zero time-varying variable, then $V(t)$, $\varpi(t)$, and $\int_0^t g(\tau)N(\varpi)\dot{\varpi} d\varpi$ must be bounded on $[0, t_f]$.

Lemma 3 *[36]:* If the solution of the resulting closed-loop system remains bounded on the interval $(0, t_f)$ for any given positive constant t, then it implies $t > 0$.

The synchronization error of follower i can be represented as

$$z'_{i,1} = \sum_{m=1}^{N_i} b_{i,m}(y_i - y_m) + b_i(y_i - y_d) \tag{4}$$

where $i = 1, \ldots, N_i$.

Lemma 4 *[37]:* Define the synchronization error vector $z_1 = (z_{1,1}, \ldots, z_{N,1})^T$, $y = (y_1, y_2, \ldots, y_N)^T$, and $\underline{y}_d = (y_d)_N^T$, then

$$\left\| y - \underline{y}_d \right\| \leq \|z_1\| / \sigma(L + B), \tag{5}$$

where $\sigma(L + B)$ is the minimum singular value of the matrix $L + B$.

Lemma 5 *[38]:* For any positive constant $k_{i,b} \in R$ in the interval $|z_{i,k}| < k_{i,b}$, we can get

$$\log \frac{k_{i,b}^2}{k_{i,b}^2 - z_{i,m}^2} \leq \frac{k_{i,b}^2}{k_{i,b}^2 - z_{i,m}^2}, \tag{6}$$

where $i = 1, \ldots, N$ and $m = 1, \ldots, l$.

C. DoS Attack Model

For the researched MASs (1), the transmission of data depends on the network channel between agents, where the network channel between agents is described as edge in the graph theory. In the unprotected network, the DoS attacks launched by malicious attackers will block the transmission of data between agents, i.e. the edge between agents is cut off. The vivid description is shown in Fig. 1. In practical application, the attackers that subjects to resource constraints can not continue to attack agents. Therefore, the intermittent DoS attack is considered in this article.

For the intermittent DoS attack [39], an instantly attack time of the edge $(i, m) \in E$ is defined as $t_{p(i,m)}$, $p \in N$. The pth DoS attack duration time is described as $T_{p(i,m)} := [t_{p(i,m)}, t_{p(i,m)} + \tau_{p(i,m)})$, where the transmission of data between follower j and follower i is blocked by attackers. $\Xi(\tau, t)$ and $\Lambda(\tau, t)$ denote the set of blocked communication channel and the set of steady communication channel in the the set of time instants (τ, t), where $\Xi(\tau, t) := \cup_{p \in N} T_{p(i,m)}$ and $\Lambda(\tau, t) := (\tau, t)/\Xi(\tau, t)$. Meanwhile, there are the following assumptions.

Assumption 3 *[39]:* Given scalars $\Theta > 0$ and $\tau_D > 0$, such that

$$n(\tau, t) \leq \Theta + \frac{t - \tau}{\tau_D}, \tag{7}$$

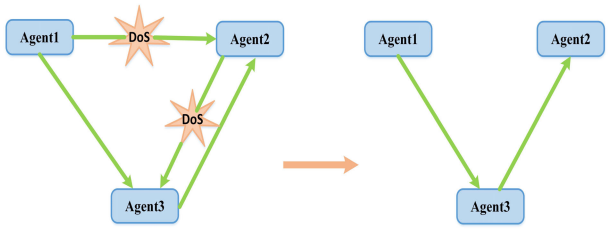


Fig. 1. The framework of DoS attacks in this paper.

where $n(\tau, t)$ denotes the number of attacks that occurred.

Assumption 4 *[39]:* Given a scalar $T \geq 1$, such that

$$|\Xi(\tau, t)| \leq \frac{t - \tau}{T}. \tag{8}$$

where $|\Xi(\tau, t)|$ denotes the attack duration.

Assumption 5: we assume that at least one communication channel attack failed.

To detect the states of the communication link between the ith agent and the mth agent, a detector $o(i, m, t)$ is set. When the edge $o(i, m, t)$ is blocked, the detector $o(i, m, t) = 1$. Otherwise, the detector $o(i, m, t) = 0$. Thus, we can obtain

$$b_{i,m}^d = \begin{cases} b_{i,m}, & o(i, m, t) = 0, \\ 0, & o(i, m, t) = 1. \end{cases} \tag{9}$$

Remark 1: Due to the DoS attacks have the characteristics of simple implementation, low cost and great impact on system stability, the DoS attacks have increased dramatically and become the main way for malicious attackers to attack systems [40]. For instance, the mobile robot systems suffer the DoS attack, the formation control performance cannot be guaranteed, even the hardware may be damaged [41]. In the industrial system control, some equipment resources are very limited. Once the industrial system is subject to the DoS attacks, the computing resources of the equipment will be exhausted quickly [42]. Therefore, the DoS attacks caught our attention.

Remark 2: Assumption 5 is reasonable for ensuring the controllability of the systems under DoS attacks, which can be found in [18] and [43]. The spanning tree is a prerequisite for achieving consensus tracking control of the MASs. If the spanning tree is not guaranteed, it is impossible to construct the controller $u(t)$ of follower to restore ideal system performance and stabilize the system. Moreover, besides making control issues meaningful, DoS attacks have practical motives to limit communication links. In the practical scenario, attackers may ensure the concealment of their attacks by deliberately allowing at least one communication link to fail for a certain period.

D. Actuator Faults

In this paper, the two types of fault are considered as lock-in-place and loss of effectiveness [44]. For the actuator, the faults model is constructed by

$$u_i^f = \rho_i u_i(t) + \tau_i, \tag{10}$$

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where ρ_i denotes the effective ratio after the actuator loses a certain efficiency and $0 < \rho_i \leq 1$, u_i is the normal control law, τ_i represents the unknown lock-in-place fault parameter, $\forall i = 1, \ldots, N$.

E. Fuzzy Logic Systems

The unknown nonlinear functions present in the system (1) are processed by fuzzy approximation technique. According to the previous works [45], we can obtain

$$\sup_{x\in\Delta}\left|\psi(x) - \theta^T S(x)\right| \leq \xi, \tag{11}$$

where $\psi(x)$ denotes the unknown smooth function that is defined on a set Δ, ξ denotes the desired level of accuracy, $\theta^T = [\theta_1, \ldots, \theta_l]$ denotes the weight vector and $S(x) = [S_1(x), \ldots, S_l(x)]^T / \sum_{k=1}^{l} S_k(x)$ denotes the basis function vector with the number of fuzzy rules l. Furthermore, the Gaussian function $S_k(x)$ is usually selected as

$$S_k(x) = \exp[\frac{-(x-\vartheta_k)^T(x-\vartheta_k)}{\gamma_k^2}] \tag{12}$$

where ϑ_k and γ_k represent the center and width of $S_k(x)$, respectively.

Remark 3: Assumption 1 is a general assumption about the leader signal y_d. Assumption 2 is usually used to deal with unknown nonlinear function. Assumption 3 and Assumption 4 imply that the DoS attacks are resources limited. Assumption 5 indicates that the communication topology of the system (1) under the DoS attacks still contains a spanning tree.

Remark 4: Lemma 1, Lemma 2 and Lemma 3 are the fundamental lemma of Nussbaum functions, which are used to deal with actuator coefficient and analysis the stability of the system (1). Lemma 4 is applied to analysis the consensus asymptotic tracking performance. Lemma 5 is used to deal with barrier Lyapunov functions.

III. Main Result

In this part, by using the backstepping method and the fuzzy state estimator, a novel distributed consensus asymptotic tracking control algorithm will be proposed for the researched system (1).

A. Fuzzy State Estimator

According to FLSs, the system (1) is reconstructed as

$$\begin{cases} \dot{x}_{i,m} = x_{i,m+1} + \theta_{i,m}^T S(\hat{\bar{x}}_{i,m}) + \Delta S_{i,m} + \sigma_{i,m}, \\ \dot{x}_{i,l} = u_i^f + \theta_{i,l}^T S(\bar{x}_{i,l}) + \Delta S_{i,l} + \sigma_{i,l}, \\ y_i = x_{i,1}, \end{cases} \tag{13}$$

where $\Delta S_{i,m} = \theta_{i,m}^T S(\bar{x}_{i,m}) - \theta_{i,m}^T S(\hat{\bar{x}}_{i,m})$, $\hat{\bar{x}}_{i,m} = [\hat{x}_{i,1}, \cdots, \hat{x}_{i,m}]^T$ is the estimation vector of $\bar{x}_{i,m}$, and $\sigma_{i,m}$ is the approximation error which satisfies $\left|\sigma_{i,m}\right| < \xi_{i,m}$. Based on the reconstructed system (14), the state estimator is costructed as

$$\begin{cases} \dot{\hat{x}}_{i,m} = A_i \hat{x}_i + K_i y_i + I_i u_i^f + \sum_{m=1}^{l} B_{i,m}[\hat{\theta}_{i,m}^T S(\hat{\bar{x}}_{i,m})], \\ \hat{y}_i = C_i^T \hat{x}_{i,1}, \end{cases} \tag{14}$$

where $A_i = \begin{pmatrix} -K_{i,1} & 1 & 0 & \cdots & 0 \\ \vdots & & I & & \vdots \\ -K_{i,l} & 0 & \cdots & \cdots & 0 \end{pmatrix}$ is a strict Hurwitz matrix, $K_i = [K_{i,1}, \cdots, K_{i,l}]^T$, $I_i = [0, \cdots, 1]^T$, $B_{i,m} = \left[0, \cdots, \underbrace{1}_{m}, \cdots, 0\right]^T$, $C_i^T = [1, 0, \cdots, 0]$, and $\hat{\theta}_i = [\hat{\theta}_{i,1}, \cdots, \hat{\theta}_{i,l}]^T$ is the estimate of θ_i. The fuzzy estimator error is defined as $e_i = x_i - \hat{x}_i, \forall i = 1, \ldots, N$. Then, from (12) and (13), one has

$$\begin{cases} \dot{e}_i = A_i e_i + \sum_{m=1}^{l} B_{i,m}(\tilde{\theta}_{i,m}^T S(\hat{\bar{x}}_{i,m}) + \Delta S_{i,m} + \sigma_{i,m}), \\ \hat{y}_i = C_i^T \hat{x}_i. \end{cases} \tag{15}$$

According to the state estimator (13), the Lyapunov function is chosen as

$$V_{i,0} = {e_i}^T M_i e_i, \tag{16}$$

meanwhile, the differential of $V_{i,0}$ is calculated as

$$\dot{V}_{i,0} = -e_i^T M_i e_i + 2e_i^T M_i\{\sum_{m=1}^{l} I_{i,m}[\tilde{\theta}_{i,m}^T S(\hat{\bar{x}}_{i,m}) + \Delta S_{i,m} + \sigma_{i,m}]\}. \tag{17}$$

There exist the matrixes $M_i^T = M_i > 0$ and $O_i^T = O_i > 0$ such that $A_i^T M_i + M_i A_i = -O_i$. According to Assumption 2, the following inequalities can be obtained:

$$2e_i^T M_i \sum_{m=1}^{l} I_{i,m}\left[\tilde{\theta}_{i,m}^T S(\hat{\bar{x}}_{i,m}) + \sigma_{i,m}\right] \leq 2le_i^T e_i + \|M_i\|^2 \sum_{m=1}^{l} (\tilde{\theta}_{i,m}^T \tilde{\theta}_{i,m} + \xi_{i,m}^2), \tag{18}$$

$$2e_i^T M_i \sum_{m=1}^{l} I_{i,m} \Delta S_{i,m} \leq (l + \|M_i\|^2 \sum_{m=1}^{l} L_{i,m}^2) e_i^T e_i. \tag{19}$$

Substituting (19)-(20) into (18) yields

$$\dot{V}_{i,0} \leq -\gamma_i e_i^T e_i + \|M_i\|^2 \sum_{m=1}^{l} (\tilde{\theta}_{i,m}^T \tilde{\theta}_{i,m} + \xi_{i,m}^2), \tag{20}$$

where $\gamma_i = \lambda_{\min}(O_i) - 3l - \|M_i\|^2 \sum_{m=1}^{l} L_{i,m}^2$.

Remark 5: To satisfy $\gamma_i > 0$, we should choose the appropriate design parameters O_i, M_i and $L_{i,m}$.

B. Event-Triggered Secure Controller Design

In this part, the DSC method is integrated into the traditional backstepping process in this paper. According to (4) and (9), we construct the coordinates transform as follows:

$$\begin{cases} z_{i,1} = \sum_{m=1}^{N_i} b_{i,m}^d (y_i - y_m) + b_i(y_i - y_d), \\ z_{i,j} = \hat{x}_{i,j} - \alpha_{i,j}^C, \\ r_{i,j} = \alpha_{i,j}^C - \alpha_{i,j-1}, \end{cases} \tag{21}$$

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where $z_{i,j}$, $\alpha_{i,j}^{C}$, $\alpha_{i,j-1}$, and $r_{i,j}$ represent the error surface, the output of a low-pass filter, the virtual control law, and the error between virtual control law and the output of a low-pass filiter, $\forall j = 1, \ldots, l$.

Step 1*:* According to (1) and (21), the synchronization error $z_{i,1}$ is calculated as

$$\dot{z}_{i,1} = (\sum_{m=1}^{N_i} b_{i,m}^{d} + b_i)(f_{i,1}(x_{i,1}) + x_{i,2}) - \sum_{m=1}^{N_i} b_{i,m}^{d}(f_{m,1}(x_{m,1}) + x_{m,2}) - b_i \dot{y}_d. \quad (22)$$

To simplify, $\sum_{m=1}^{N_i} b_{i,m}^{d} + b_i$ is defined as φ_i. The BLF candidate is given as follows:

$$V_{i,1} = \frac{1}{2}\log\frac{k_{i,b_1}^2}{k_{i,b_1}^2 - z_{i,1}^2} + \frac{1}{2l_{i,1}}\tilde{\theta}_{i,1}^{T}\tilde{\theta}_{i,1} + \frac{1}{2\beta_{i,1}}\tilde{\omega}_{i,1}^2 + \frac{1}{2}r_{i,2}^2, \quad (23)$$

where $l_{i,1}$ and $\beta_{i,1}$ represent the design constants, $\log(\cdot)$ represents the natural logarithm function, $\tilde{\theta}_{i,1} = \theta_{i,1} - \hat{\theta}_{i,1}$ and $\tilde{\omega}_{i,1} = \omega_{i,1} - \hat{\omega}_{i,1}$.

In the compact set $\Omega_{z_{i,1}} = \{z_{i,1} : |z_{i,1}| < k_{i,b_1}\}$, $V_{i,1}$ is continuous. Further, one has

$$\dot{V}_{i,1} = \Delta_{i,1}[\varphi_i(f_{i,1}(x_{i,1}) + e_{i,2} + z_{i,2} + r_{i,2} + \alpha_{i,1}) - \sum_{m=1}^{N_i} b_{i,m}^{d}(f_{m,1}(x_{m,1}) + \hat{x}_{m,2} + e_{m,2}) - b_i \dot{y}_d] - \frac{1}{l_{i,1}}\tilde{\theta}_{i,1}^{T}\dot{\hat{\theta}}_{i,1} - \frac{1}{\beta_{i,1}}\tilde{\omega}_{i,1}\dot{\hat{\omega}}_{i,1} + r_{i,2}\dot{r}_{i,2}, \quad (24)$$

where $\Delta_{i,1} = z_{i,1}/(k_{i,b_1}^2 - z_{i,1}^2)$. Applying Young's inequality, one gets

$$\Delta_{i,1}\varphi_i(e_{i,2} + z_{i,2} + r_{i,2}) \le \frac{3}{2}\Delta_{i,1}^2 + \frac{{\varphi_i}^2}{2}{e_i}^T e_i + \frac{{\varphi_i}^2}{2}z_{i,2}^2 + \frac{{\varphi_i}^2}{2}r_{i,2}^2, \quad (25)$$

$$-\Delta_{i,1}\sum_{m=1}^{N_i} b_{i,m}^{d} e_{m,2} \le \frac{N_i}{2}\Delta_{i,1}^2 + \frac{\sum_{m=1}^{N}(b_{i,m}^{d})^2}{2}e_{m,2}^2. \quad (26)$$

Considering (24) and (25), it follows that

$$\dot{V}_{i,1} \le \Delta_{i,1}[\varphi_i\theta_{i,1}^{T}S_{i,1}(x_{i,1}) + \varphi_i\alpha_{i,1} + \frac{3}{2}\varphi_i^2\Delta_{i,1} + \frac{N_i}{2}\Delta_{i,1} + \bar{F}_{i,1}(\sigma_{i,1})] + \frac{\varphi_i^2}{2}\|e_i\|^2 + \frac{\varphi_i^2}{2}z_{i,2}^2 + \frac{\varphi_i^2}{2}r_{i,2}^2 - \frac{1}{l_{i,1}}\tilde{\theta}_{i,1}^{T}\hat{\theta}_{i,1} - \frac{1}{\beta_{i,1}}\tilde{\omega}_{i,1}^{T}\hat{\omega}_{i,1} + \frac{\sum_{m=1}^{N_i}(b_{i,m}^{d})^2}{2}e_{m,2}^2 + r_{i,2}\dot{r}_{i,2}, \quad (27)$$

where

$$\bar{F}_{i,1}(\sigma_{i,1}) = -\sum_{m=1}^{N_i} b_{i,m}^{d}\hat{x}_{m,2} - \sum_{m=1}^{N_i} b_{i,m}^{d} f_{m,1}(x_{m,1}) - b_i\dot{y}_d. \quad (28)$$

To compensate the unknown nonlinearity $\bar{F}_{i,1}(\sigma_{i,1})$, the FLS $\omega_{i,1}^{T}\bar{S}(\sigma_{i,1})$ is used as the approximator, and then, one has

$$\bar{F}_{i,1}(\sigma_{i,1}) = \theta_{i,1}^{*T}\bar{S}_{i,1}(\sigma_{i,1}) + \varepsilon_{i,1}(\sigma_{i,1}), |\varepsilon_{i,1}(\sigma_{i,1})| < \xi_{i,1}, \quad (29)$$

where $\varepsilon_{i,1}(\sigma_{i,1})$ denotes the compensating error of FLS and $\sigma_{i,1} = (x_{1,1}, \ldots, x_{N,1}, \hat{x}_{1,2}, \ldots, \hat{x}_{N,2}, \dot{y}_d)^T$. Furthermore, applying Young's inequality, one gets

$$-\Delta_i\theta_{i,1}^{*T}\bar{S}_{i,1}(\sigma_{i,1}) \le \frac{E_{i,1}}{2}\Delta_i^2\omega_{i,1}\bar{S}_{i,1}^{T}(\sigma_{i,1})\bar{S}_{i,1}(\sigma_{i,1}) + \frac{1}{2E_{i,1}}, \quad (30)$$

$$-\Delta_{i,1}\varepsilon_{i,1}(\sigma_{i,1}) \le \frac{1}{2}\Delta_{i,1}^2 + \frac{1}{2}\xi_{i,1}^2. \quad (31)$$

where $\omega_{i,1} = \| \theta_{i,1} \|^2$. Based on the previous inequalities, one has

$$\alpha_{i,1} = N(\varpi_{i,1})\bar{\alpha}_{i,1}, \quad (32)$$

$$\bar{\alpha}_{i,1} = C_{i,1}z_{i,1} + \frac{4 + N_i}{2}\Delta_{i,1} + \hat{\theta}_{i,1}S_{i,1}(x_{i,1}) + \frac{E_{i,1}}{2}\hat{\omega}_{i,1}\bar{S}_{i,1}^{T}(\sigma_{i,1})\bar{S}_{i,1}(\sigma_{i,1})\Delta_i, \quad (33)$$

where $C_{i,1}$, $P_{i,1}$, and $E_{i,1}$ are defined as the positive constants.

The control laws of adaptive parameters $\omega_{i,1}$ and $\theta_{i,1}$ are constructed as

$$\dot{\hat{\theta}}_{i,1} = l_{i,1}\varphi_i S_{i,1}(x_{i,1})\Delta_{i,1} - h_{i,1}\hat{\theta}_{i,1}, \quad (34)$$

$$\dot{\hat{\omega}}_{i,1} = \frac{E_{i,1}}{2}\Delta_i^2\bar{S}_{i,1}^{T}(\sigma_{i,1})\bar{S}_{i,1}(\sigma_{i,1}) - \bar{h}_{i,1}\hat{\omega}_{i,1}, \quad (35)$$

where both $h_{i,1}$ and $\bar{h}_{i,1}$ denote the tunable constants, $l_{i,1}$ and $\beta_{i,1}$ are given in (22). When $\alpha_{i,1}$ pass through a low-pass filter with time positive constant $o_{i,2}$, we have

$$o_{i,2}\dot{\alpha}_{i,2}^{C} + \alpha_{i,2}^{C} = \alpha_{i,1}, \alpha_{i,2}^{C}(0) = \alpha_{i,1}(0). \quad (36)$$

where $o_{i,2}$ is the design parameter. Then, we have

$$\left|\dot{r}_{i,2} + \frac{r_{i,2}}{o_{i,2}}\right| \le \Phi_{i,2}(z_{i,1}, z_{i,2}, \hat{\theta}_{i,1}, \hat{\omega}_{i,1}, y_d, \dot{y}_d, \ddot{y}_d), \quad (37)$$

where $\Phi_{i,2}$ is a continuous function and $\Phi_{i,2} < \Gamma_{i,2}$.

Remark 6: For more information about the formula (37) and $\Gamma_{i,2}$, see [46] and [47].

According to (37), one has

$$r_{i,2}\dot{r}_{i,2} \le |r_{i,2}|\Phi_{i,2} - \frac{r_{i,2}^2}{o_{i,2}} \le -\frac{r_{i,2}^2}{o_{i,2}} + \frac{1}{2}r_{i,2}^2 + \frac{1}{2}\Gamma_{i,2}^2. \quad (38)$$

Combining (30)-(35), (38) and Lemma 5, (27) is rewritten as

$$\dot{V}_{i,1} \le -C_{i,1}\log\frac{k_{i,b_1}^2}{k_{i,b_1}^2 - z_{i,1}^2} + \frac{h_{i,1}}{l_{i,1}}\tilde{\theta}_{i,1}^{T}\hat{\theta}_{i,1} + \frac{\bar{h}_{i,1}}{\beta_{i,1}}\tilde{\omega}_{i,1}\hat{\omega}_{i,1} + (\frac{\varphi_{i,1}^2 + 1}{2} - \frac{1}{o_{i,2}})r_{i,2}^2 + \Delta_{i,1}\bar{\alpha}_{i,1} + \varphi_i N(\varpi_{i,1})\Delta_{i,1}\bar{\alpha}_{i,1} + \frac{\varphi_{i,1}^2}{2}z_{i,2}^2 + \frac{1}{2}\|e_i\|^2 + \frac{\sum_{m=1}^{N_i}(b_{i,m}^{d})^2}{2}e_{k,2}^2 + \zeta_{i,1}, \quad (39)$$

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where $\zeta_{i,1} = \frac{1}{2}\Gamma_{i,2}^2 + \frac{1}{2}\xi_{i,1}^2 + \frac{1}{2E_{i,1}^2}$. It is point out that these items $\Delta_{i,1}\bar{\alpha}_{i,1}$, $\varphi_i\Delta_{i,1}N(\varpi_{i,1})\bar{\alpha}_{i,1}$, $\frac{1}{2}z_{i,2}^2$, $\frac{h_{i,1}}{l_{i,1}}\tilde{\theta}_{i,1}^T\hat{\theta}_{i,1}$, and $\frac{\bar{h}_{i,1}}{l_{i,1}}\tilde{\omega}_{i,1}^T\hat{\omega}_{i,1}$ will be handled later.

Step $k(k = 2, 3, \ldots, l-1)$*:* The BLF candidate is given as follows:

$$V_{i,k} = V_{i,k-1} + \frac{1}{2}\log\frac{k_{i,b_k}^2}{k_{i,b_k}^2 - z_{i,k}^2} + \frac{1}{2l_{i,k}}\tilde{\theta}_{i,k}^T\tilde{\theta}_{i,k} + \frac{1}{2}r_{i,k+1}^2, \tag{40}$$

where $l_{i,k}$ is a design parameter, $l_{i,k} > 0$. Considering (13) and (20), the derivative of $V_{i,k}$ yields

$$\begin{aligned}\dot{V}_{i,k} = \dot{V}_{i,k-1} + \Delta_{i,k}(R_{i,k}e_{i,1} + z_{i,k+1} + r_{i,k+1} + \alpha_{i,k} \\ + \hat{\theta}_{i,k}^T S(\hat{\bar{x}}_{i,k}) - \dot{\alpha}_{i,k}^C + \tilde{\theta}_{i,k}^T S(\hat{\bar{x}}_{i,k}) - \tilde{\theta}_{i,k}^T S(\hat{\bar{x}}_{i,k})) \\ - \frac{1}{l_{i,k}}\tilde{\theta}_{i,k}^T\hat{\theta}_{i,k} + r_{i,k+1}\dot{r}_{i,k+1},\end{aligned} \tag{41}$$

where $\Delta_{i,k} = z_{i,k}/(k_{i,b_k}^2 - z_{i,k}^2)$. Using Young's inequality, one has $\Delta_{i,k}(z_{i,k+1} + r_{i,k+1}) \le \Delta_{i,k}^2 + \frac{1}{2}z_{i,k+1}^2 + \frac{1}{2}r_{i,k+1}^2$ and $-\Delta_{i,k}\tilde{\theta}_{i,k}^T S(\bar{x}_{i,k}) \le \frac{1}{2}\Delta_{i,k}^2 + \frac{1}{2}\tilde{\theta}_{i,k}^T\tilde{\theta}_{i,k}$.

Then, the virtual control $\alpha_{i,k}$ in (20) is developed as

$$\alpha_{i,k} = N(\varpi_{i,k})\bar{\alpha}_{i,k}, \tag{42}$$

$$\begin{aligned}\bar{\alpha}_{i,k} = C_{i,k}z_{i,k} + \frac{3}{2}\Delta_{i,k} + K_{i,k}e_{i,1} \\ + \hat{\theta}_{i,k}S_{i,k}(x_{i,k}) + \dot{\alpha}_{i,k}^C,\end{aligned} \tag{43}$$

where $C_{i,k}$ and $E_{i,1}$ are the tunable parameters. The adaptive laws for $\hat{\theta}_{i,k}$ is constructed as

$$\dot{\hat{\theta}}_{i,k} = l_{i,k}\Delta_{i,k}S(\hat{\bar{x}}_{i,k}) - h_{i,k}\hat{\theta}_{i,k}, \tag{44}$$

where $h_{i,k}$ is the design parameter and $l_{i,k}$ is defined in (40).

When the low-pass filter with the positive constant is applied, we have

$$o_{i,k}\dot{\alpha}_{i,k+1}^C + \alpha_{i,k+1}^C = \alpha_{i,k}, \alpha_{i,k+1}^C(0) = \alpha_{i,k}(0). \tag{45}$$

Then, we have

$$\begin{aligned}\left|\dot{r}_{i,k+1} + \frac{r_{i,k+1}}{o_{i,k+1}}\right| \le \Phi_{i,k+1}(z_{i,1}, \ldots, z_{i,k+1}, \\ \hat{\theta}_{i,1}, \ldots, \hat{\theta}_{i,k}, y_d, \dot{y}_d, \ddot{y}_d),\end{aligned} \tag{46}$$

where $\Phi_{i,k+1}$ is a continuous function and $\Phi_{i,k+1} < \Gamma_{i,k+1}$.

According to (44), one has

$$\begin{aligned}r_{i,k+1}\dot{r}_{i,k+1} &\le |r_{i,k+1}|\Phi_{i,k+1} - \frac{r_{i,k+1}^2}{o_{i,k+1}} \\ &\le -\frac{r_{i,k+1}^2}{o_{i,k+1}} + \frac{1}{2}r_{i,k+1}^2 + \frac{1}{2}\Gamma_{i,k+1}^2.\end{aligned} \tag{47}$$

From (42)-(44) and (47), (41) is changed as

$$\begin{aligned}\dot{V}_{i,k} \le -\sum_{m=1}^{k} C_{i,m}\log\frac{k_{i,b_m}^2}{k_{i,b_m}^2 - z_{i,m}^2} + \sum_{m=1}^{k}\frac{h_{i,m}}{l_{i,m}}\tilde{\theta}_{i,m}^T\hat{\theta}_{i,m} \\ + \frac{\bar{h}_{i,1}}{\beta_{i,1}}\tilde{\omega}_{i,1}^T\hat{\omega}_{i,1} + \sum_{m=1}^{k}(1 - \frac{1}{o_{i,m+1}})r_{i,m+1}^2 \\ + \sum_{m=1}^{k}\Delta_{i,m}\bar{\alpha}_{i,m} + \varphi_i N(\varpi_{i,1})\Delta_{i,1}\bar{\alpha}_{i,1} \\ + \sum_{m=2}^{k} N(\varpi_{i,m})\Delta_{i,m}\bar{\alpha}_{i,m} + \frac{1}{2}z_{i,k+1}^2 + \frac{1}{2}\|e_i\|^2 \\ + \frac{\sum_{m=1}^{N_i}(b_{i,m}^d)^2}{2}e_{m,2}^2 + \sum_{m=2}^{k}\frac{1}{2}\tilde{\theta}_{i,m}^T\tilde{\theta}_{i,m} + \sum_{m=1}^{k}\zeta_{i,m}.\end{aligned} \tag{48}$$

Step l: Similar to Step k, the BLF candidate is given as follows:

$$V_{i,l} = V_{i,l-1} + \frac{1}{2}\log\frac{k_{i,b_l}^2}{k_{i,b_l}^2 - z_{i,l}^2} + \frac{1}{2l_{i,l}}\tilde{\theta}_{i,l}^T\tilde{\theta}_{i,l} + \frac{1}{2}\tau_i^2, \tag{49}$$

where $l_{i,l}$ is a turable parameter, $l_{i,l} > 0$. Then, From (13), (20) and (49), the derivative of $V_{i,l}$ yields

$$\begin{aligned}\dot{V}_{i,l} = \dot{V}_{i,l-1} + \Delta_{i,k}(R_{i,k}e_{i,1} + \rho_i u_i + \tau_i + \hat{\theta}_{i,l}^T S(\hat{\bar{x}}_{i,l}) \\ - \dot{\alpha}_{i,l}^C + \tilde{\theta}_{i,l}^T S(\hat{\bar{x}}_{i,l}) - \tilde{\theta}_{i,k}^T S(\hat{\bar{x}}_{i,l})) - \frac{1}{l_{i,k}}\tilde{\theta}_{i,l}^T\hat{\theta}_{i,l} \\ - \tilde{\tau}_i\dot{\hat{\tau}}_i,\end{aligned} \tag{50}$$

where $\Delta_{i,l}=z_{i,l}/(k_{i,b_l}^2 - z_{i,l}^2)$. Further, we have $-\Delta_{i,l}\tilde{\theta}_{i,l}^T S(\hat{\bar{x}}_{i,l}) \le \frac{1}{2}\Delta_{i,l}^2 + \frac{1}{2}\tilde{\theta}_{i,l}^T\tilde{\theta}_{i,l}$. Then, the virtual control $\alpha_{i,l}$ in (20) is developed as

$$\alpha_{i,l} = N(\varpi_{i,l})\bar{\alpha}_{i,l}, \tag{51}$$

$$\begin{aligned}\bar{\alpha}_{i,l} = C_{i,l}z_{i,l} + \frac{1}{2}\Delta_{i,l} + K_{i,l}e_{i,1} \\ + \hat{\theta}_{i,l}S_{i,l}(\hat{\bar{x}}_{i,l}) + \dot{\alpha}_{i,l}^C + \hat{\tau}_i,\end{aligned} \tag{52}$$

where $C_{i,l}$ and Q_i are the turable parameters.

To reduce the communication burden of the proposed controller, we given the following relative threshold control strategy:

$$\begin{aligned}&\bar{u}_i(t) = -(1 + \bar{\delta}_i)[\alpha_{i,l}\tanh(\frac{\Delta_{i,l}\alpha_{i,l}}{\bar{\omega}_i}) + \bar{m}_i\tanh(\frac{\Delta_{i,l}\bar{m}_i}{\bar{\omega}_i})], \\ &u_i(t) = \bar{u}_i(t_m), \forall t \in [t_m, t_{m+1}), \\ &t_{m+1} = \inf\{t \in R||e_i^c(t)| \ge \bar{\delta}_i|u_i(t)| + \vartheta_i\},\end{aligned}$$

where $e_i^c = u_i(t) - \bar{u}_i(t)$, $0 < \bar{\delta}_i < 1$, $\vartheta_i > 0$, $\bar{m}_i > \vartheta_i/(1-\bar{\delta}_i)$ and $\bar{\omega}_i > 0$ are the designed parameters. Based on the above consideration, one has

$$\bar{u}_i(t) = (1 + \rho_{i,1}(t)\bar{\delta}_i)u_i(t) + \rho_{i,2}(t), t \in [t_k, t_{k+1}),$$

where $|\rho_{i,1}(t)| \le 1$ and $|\rho_{i,2}(t)| \le 1$ are the time-varying parameters. Therefore, the controller can be obtained as

$$u_i(t) = \frac{\bar{u}_i - \rho_{i,2}\vartheta_i}{1 + \rho_{i,1}\bar{\delta}_i}. \tag{53}$$

The adaptive laws for $\hat{\theta}_{i,l}$ and $\hat{\tau}_{i,l}$ are constructed as

$$\dot{\hat{\theta}}_{i,l} = l_{i,l}\Delta_{i,l}S(\hat{\bar{x}}_{i,l}) - h_{i,l}\hat{\theta}_{i,l}, \tag{54}$$

$$\dot{\hat{\tau}}_{i,l} = \Delta_{i,l} - \hat{\tau}_{i,l}. \tag{55}$$

Substituting (52)-(55) into (50) yields

$$\begin{aligned}\dot{V}_{i,l} \le \dot{V}_{i,l-1} + \Delta_{i,l}\rho_i[(\frac{\bar{u}_i - \rho_{i,2}\vartheta_i}{1 + \rho_{i,1}\bar{\delta}_i}) - N(\varpi_{i,l})\bar{\alpha}_{i,l}] \\ + \rho_i N(\varpi_{i,l})\Delta_{i,l}\bar{\alpha}_{i,l} + \Delta_{i,l}\bar{\alpha}_{i,l} + \frac{h_{i,l}+1}{l_{i,l}}\tilde{\theta}_{i,l}^T\hat{\theta}_{i,l}\end{aligned}$$

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$$+\tilde{\tau}_{i,l}\hat{\tau}_{i,l}+\zeta_{i,l}-\frac{1}{2}z_{i,l}^2, \tag{56}$$

where $\zeta_{i,l}=\frac{1}{2E_{i,l}}+\frac{1}{2Q_i}$. Then, we can yield

$$\frac{\Delta_{i,l}\rho_i\bar{u}_i}{1+\rho_{i,1}\bar{\delta}_i}\le\frac{\Delta_{i,l}\rho_i\bar{u}_i}{1+\bar{\delta}_i}. \tag{57}$$

It is worth noting that the hyperbolic tangent function $\tanh(\cdot)$ has the following property [48]:

$$0\le|\psi|-\psi\tanh(\frac{\psi}{\upsilon})\le 0.2785\upsilon, \tag{58}$$

where $\upsilon>0$, $\psi\in S$. Substituting (58) into (56) gives

$$\begin{aligned}
&\Delta_{i,l}\rho_i[(\frac{\bar{u}_i-\rho_{i,2}\vartheta_i}{1+\rho_{i,1}\bar{\delta}_i})-N(\varpi_{i,l})\bar{\alpha}_{i,l}]\\
&\le\Delta_{i,l}\rho_i\frac{\bar{u}_i}{1+\bar{\delta}_i}+|\frac{\Delta_{i,l}\rho\vartheta_i}{1-\bar{\delta}_i}|-N(\varpi_{i,l})\bar{\alpha}_{i,l}\\
&\le-\Delta_{i,l}\rho_i\alpha_{i,l}\tanh(\frac{\Delta_{i,l}\alpha_{i,l}}{\bar{\omega}_i})+|\frac{\Delta_{i,l}\rho\vartheta_i}{1-\bar{\delta}_i}|\\
&\quad-\Delta_{i,l}\rho_i\bar{m}_i\tanh(\frac{\Delta_{i,l}\bar{m}_i}{\bar{\omega}_i})-N(\varpi_{i,l})\bar{\alpha}_{i,l}\\
&\le-|\Delta_{i,l}\alpha_{i,l}|+0.557\rho_i\upsilon_i-N(\varpi_{i,l})\bar{\alpha}_{i,l}\\
&\le 0.557\upsilon_i.
\end{aligned} \tag{59}$$

By Lemma 5 and (59), it is obtained that

$$\begin{aligned}
\dot{V}_{i,l}\le&-\sum_{m=1}^{l}C_{i,m}\log\frac{k_{i,b_m}^2}{k_{i,b_m}^2-z_{i,m}^2}+\sum_{m=1}^{l}\frac{h_{i,m}}{l_{i,m}}\tilde{\theta}_{i,m}^T\hat{\theta}_{i,m}\\
&+\frac{\bar{h}_{i,1}}{\beta_{i,1}}\tilde{\omega}_{i,1}^T\hat{\omega}_{i,1}+\tilde{\tau}_{i,l}\hat{\tau}_{i,l}+\sum_{m=1}^{l-1}(1-\frac{1}{o_{i,m+1}})r_{i,m+1}^2\\
&+\sum_{m=1}^{l}\Delta_{i,m}\bar{\alpha}_{i,m}+\varphi_iN(\varpi_{i,1})\Delta_{i,1}\bar{\alpha}_{i,1}\\
&+\sum_{m=2}^{l-1}N(\varpi_{i,m})\Delta_{i,m}\bar{\alpha}_{i,m}+\rho_iN(\varpi_{i,l})\Delta_{i,l}\bar{\alpha}_{i,l}\\
&+\sum_{m=2}^{l}\frac{1}{2}\tilde{\theta}_{i,m}^T\tilde{\theta}_{i,m}+\frac{1}{2}\|e_i\|^2+\frac{\sum_{m=1}^{N_i}(b_{i,m}^d)^2}{2}e_{m,2}^2\\
&+\sum_{m=1}^{l}\zeta_{i,m}.
\end{aligned} \tag{60}$$

IV. Stability Analysis

Based on the previous design process, the proposed security control strategy is summarized as Algorithm 1. Further, the consensus asymptotic tracking performance and the stability of the researched system (1) will be presented in the following Theorem 1.

Theorem 1: Consider the researched system (1) under the event-triggered controller (56), and the adaptive laws (34), (35), (44), (54) and (55), it is concluded that all closed-loop signals remain semiglobally bounded, while each output tracking error asymptotically converges to zero and the state constraint requirements are not violated.

Algorithm 1 Security Control Strategy

Initialization: Choose proper initial conditions.

Algorithm Process:

1: **while** $i\le N$ **do**
2: construct a state estimator (13) for the ith follwer.
3: choose appropriate matrices O_i, B_i and parameter $L_{i,m}$, $m=1,\ldots,l$.
4: **if** the detector $o(i,j,t)=0$ **then**
the weight of the edge (i,j) $b_{i,j}=0$.
else
the weight of the edge (i,j) $b_{i,j}=1$.
5: design the synchronization error $z_{i,1}$ and errors $z_{i,m}$, $m=2,\ldots,l$
6: determine the virtual controls (32), (42), actual controller (57) and the adaptive laws (34), (35), (54) and (55).
7: **end while**

Proof: Construct the Lyapunov function

$$V=\sum_{i=1}^{N}(V_{i,0}+V_{i,l}). \tag{61}$$

The differential of V is calculated as

$$\begin{aligned}
\dot{V}\le&\sum_{i=1}^{N}\{-(\gamma_i-\frac{1}{2}-\frac{\sum_{m=1}^{N_i}(b_{i,m}^d)^2}{2})e_i^Te_i\\
&-\sum_{m=1}^{l}C_{i,m}\log\frac{k_{i,b_m}^2}{k_{i,b_m}^2-z_{i,m}^2}+\sum_{m=1}^{l}\frac{h_{i,m}}{l_{i,m}}\tilde{\theta}_{i,m}^T\hat{\theta}_{i,m}\\
&+\frac{\bar{h}_{i,1}}{\beta_{i,1}}\tilde{\omega}_{i,1}^T\hat{\omega}_{i,1}+\tilde{\tau}_{i,l}\hat{\tau}_{i,l}+\sum_{m=1}^{l-1}(1-\frac{1}{o_{i,m+1}})r_{i,m+1}^2\\
&+\sum_{m=1}^{l}\Delta_{i,m}\bar{\alpha}_{i,m}+\varphi_iN(\varpi_{i,1})\Delta_{i,1}\bar{\alpha}_{i,1}\\
&+\sum_{m=2}^{l-1}N(\varpi_{i,m})\Delta_{i,m}\bar{\alpha}_{i,m}+\rho_iN(\varpi_{i,l})\Delta_{i,l}\bar{\alpha}_{i,l}\\
&+\sum_{m=2}^{l}\frac{1}{2}\tilde{\theta}_{i,k}^T\tilde{\theta}_{i,k}+\|M_i\|^2\sum_{m=1}^{l}\tilde{\theta}_{i,m}^T\tilde{\theta}_{i,m}\\
&+\|M_i\|^2\sum_{m=1}^{l}\xi_{i,m}^2+\sum_{m=1}^{l}\zeta_{i,m}\}.
\end{aligned} \tag{62}$$

Further, we can

$$\frac{h_{i,m}}{l_{i,m}}\tilde{\theta}_{i,m}^T\hat{\theta}_{i,m}\le-\frac{h_{i,m}}{2l_{i,m}}\tilde{\theta}_{i,m}^T\tilde{\theta}_{i,m}+\frac{h_{i,m}}{2l_{i,m}}\theta_{i,m}^T\theta_{i,m}, \tag{63}$$

$$\frac{\bar{h}_{i,1}}{\beta_{i,1}}\tilde{\omega}_{i,1}\hat{\omega}_{i,1}\le-\frac{\bar{h}_{i,1}}{2\beta_{i,1}}\tilde{\omega}_{i,1}^2+\frac{\bar{h}_{i,1}}{2\beta_{i,1}}\omega_{i,1}^2. \tag{64}$$

Substituting (63) and (64) into (62) leads to

$$\dot{V}\le-aV+\sum_{i=1}^{N}\{B_{i,1}N(\varpi_{i,1}(t))\dot{\varpi}_{i,1}(t)$$

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$$+\sum_{m=2}^{l-1} N(\varpi_{i,m}(t))\dot{\varpi}_{i,m}(t) + B_{i,2}N(\varpi_{i,l}(t))\dot{\varpi}_{i,l}(t) + \dot{\varpi}_i(t)\} + D, \tag{65}$$

where $a = \min\{2(\gamma_i - \frac{1}{2} - \frac{\sum_{m=1}^{N}(b_{i,m}^d)^2}{2}), 2C_{i,j}, 2C_{i,l}, 2(\frac{h_{i,m}}{l_{i,m}} - \frac{1}{2} - \|M_i\|^2), \frac{2\bar{h}_{i,1}}{\beta_{i,1}}, 1, (1-\frac{1}{o_{i,m+1}})|j = 1, \ldots, l-1\}$, $B_{i,1} = \varphi_i$, $B_{i,2} = \rho_i$, $\dot{\varpi}_{i,m} = \triangle_{i,m}\bar{\alpha}_{i,m}$ and $D = \sum_{i=1}^{N}\{\|M_i\|^2\sum_{m=1}^{l}\xi_{i,m}^2 + \sum_{m=1}^{l}\zeta_{i,m} + \sum_{m=1}^{l}\frac{h_{i,m}}{2l_{i,m}}\theta_{i,m}^T\theta_{i,m} + \frac{\bar{h}_{i,1}}{2\beta_{i,1}}\omega_{i,1}^2 + \frac{1}{2}\tau_{i,l}^2\}$. What is noteworthy is that the design parameters O_i, M_i, L_i, $h_{i,m}$ and $l_{i,m}$ can be chosen such that $2(\gamma_i - \frac{1}{2} - \frac{\sum_{m=1}^{N}(b_{i,m}^d)^2}{2}) > 0$ and $2(\frac{h_{i,m}}{l_{i,m}} - \frac{1}{2} - \|M_i\|^2) > 0$ are satisfied. Therefore, Lemma 2 can be directly utilized to (65). It proves that the proposed distributed fuzzy secure consensus tracking control algorithm ensures the boundness of the researched system (1).

Combining results in (33), (43), and (52), each item of $\dot{\varpi}_{i,m}$, $m = 1, \ldots, l$, must always be non-negative for $t \geq 0$. Thus, $\sum_{m=1}^{l}\dot{\varpi}_{i,m} \geq \sum_{m=1}^{l}\frac{C_{i,m}z_{i,m}^2}{k_{i,b_m}^2 - z_{i,m}^2}$ and its time integration over $[0, t]$ leads to $\int_0^t \sum_{m=1}^{l}\frac{C_{i,m}z_{i,m}^2}{k_{i,b_m}^2 - z_{i,m}^2} \leq \sum_{m=1}^{l}\varpi_{i,m}(t) - \varpi_{i,m}(0)$, where $\varpi_{i,m}(t)$ has been proved bounded. Meanwhile, considering that $C_{i,m}$ and k_{i,b_m} are both designed parameters, one can derive that $z_{i,m}$ is square integrable. After using Lemma 1, one has

$$\lim_{t\to\infty} z_{i,m}(t) = 0, \quad i = 1, \ldots, N, m = 1, \ldots, l. \tag{66}$$

According to (20), (67), and Lemma 1, one has $\lim_{t\to\infty} y_i - y_d = 0$, which illustrates the output of the ith follower asymptotically converges to the leader signal.

The following proof shows that the state constraint cannot be violated. The synchronization error is rewritten as

$$z_1 = (L + B)(y - \mathbf{1}y_d), \tag{67}$$

where $z_1 = [z_{1,1}, \ldots, z_{N,1}]^T$, $B = diag\{b_1, \ldots, b_N\}$, $y = [y_1, \ldots, y_N]^T$, and $\mathbf{1} = [1, \ldots, 1]_{N\times 1}^T$. According to (22) and Lemma 3, one has $|z_{i,1}| < k_{i,b_1}$ and $|y_d| < y_0$. Then, one has $x_{i,1} < \sigma(L + B)^{-1}k_{i,b_1} + y_0 < k_{i,c_1}$. From (65), we have that

$$V \leq \eta + \bar{c}_0, \tag{68}$$

where $\bar{c}_0 = V(0) + D/a$, $\eta = \sum_{i=1}^{N}\{e^{-at}\int_0^t[(B_{i1}N(\varpi_{i,1}(t))\dot{\varpi}_{i,1}(t) + \sum_{m=2}^{l-1}N(\varpi_{i,m}(t))\dot{\varpi}_{i,m}(t) + B_{i,2}N(\varpi_{i,l}(t))\dot{\varpi}_{i,l}(t)] + \dot{\varpi}_i(t))e^{at}d\tau\}$ and there exists a constant $\bar{\eta} > 0$ such that $\eta \leq \bar{\eta}$. According to $z_{i,2} = \hat{x}_{i,2} - \alpha_{i,2}^C$, then $|x_{i,2}| \leq |z_{i,2}| + |e_{i,2}| + |r_{i,2}| + |\alpha_{i,1}|$. Because $\alpha_{i,1}$ is a continuous function of $x_{i,1}$ and $z_{i,1}$, we have $|x_{i,1}| \leq k_{i,c_1}$, $|z_{i,1}| < k_{i,b_1}$, then there exists a constant $\bar{\alpha}_{i,1}^* > 0$ with $|\alpha_{i,1}| < \bar{\alpha}_{i,1}^*$. From (68), we can obtain $|e_{i,2}| \leq \|e_i\| \leq \sqrt{[2/\lambda_{\min}(M_i)]}(\bar{c}_0 + \bar{\eta}) = \Psi_{i,2}$, $|r_{i,2}| \leq \|r_{i,2}, r_{i,3}, \ldots, r_{i,l}\| \leq \sqrt{2}(\bar{c}_0 + \bar{\eta}) = \Xi_{i,2}$, yielding $|x_{i,2}| \leq k_{i,b_2} + \Psi_{i,2} + \Xi_{i,2} + \bar{\alpha}_{i,1}$. Let $k_{i,b_2} = k_{c_2} - \Psi_{i,2} - \Xi_{i,2} - \bar{\alpha}_{i,1}$, we can obtain $|x_{i,2}| < k_{i,c_2}$. The proof process of the constrained $x_{i,j}$ is similar to the proof process of the constrained $x_{i,2}$, $j = 3, \ldots, l$. Hence, the proof is completed.

TABLE I
THE CONSIDERED DOS ATTACKS

Attack duration time (s)	Attacked edge
0.4 ∼ 0.6	(3,2), (5,1)
1.45 ∼ 2.15	(3,1), (5,4)
3.55 ∼ 4.05	(3,2), (5,1)
4.4 ∼ 5	(4,5), (2,3)
5.25 ∼ 5.75	(4,1), (2,1)
6.2 ∼ 6.4	(3,2), (5,1)

Remark 7: To show that the Zeno-behavior is successfully avoided, first, it is proved that there exists a $\bar{t}_f > 0$ such that $t_{k+1} - t_k > \bar{t}_f$ for $\forall k$. According to the definition of $e_i^c(t)$, $e_i^c(t) = u_i(t) - \bar{u}_i(t)$ can be obtained, then $\frac{d}{dt}|e_i^c(t)| = sign(e_i^c(t))\dot{e}_i^c(t) \leq |\dot{\bar{u}}_i(t)|$. According to the relative threshold control strategy, the functions $\alpha_{i,l}$, $\rho_{i,1}(t)$, and $\rho_{i,2}(t)$ can be obtained. Because these signals are bounded, the $\bar{u}_i(t)$ is bounded. Moreover, we have that $\bar{u}_i(t)$ is a smooth function, so $\dot{\bar{u}}_i(t)$ is bounded. There exist a constant $\bar{M} > 0$ such that $|\dot{\bar{u}}_i| \leq \bar{M}$. Then, using the mean value theorem and $e_i^c(t_k) = 0$, we can obtain $\lim_{t\to t_{k+1}} e_i^c(t_{k+1}) = \bar{\delta}_i|u_i(t)| + \vartheta_i$. Then, it can be ensured that the lower bounded of $\bar{t}_f$ is $(\bar{\delta}_i|u_i(t)| + \vartheta_i)/\bar{\delta}_i$. During the controller design process, $\bar{t}_f \neq 0$ still be ensured. Therefore, the Zeno-behavior is effectively avoided.

V. SIMULATION STUDY

In order to better discuss the validity and robustness of the proposed control algorithm, two simulation examples are presented in this section.

Example 1: To confirm that the proposed control algorithm is correct, the MASs made up of numerical system models a re taken into consideration. Fig. 2 shows the topology relationship of the seven agents in the directed graph G_1, where the agent marked 0 denotes the leader and other marked agents represent followers. In the directed graph G_1, the adjacency matrix A and the Laplacian matrix L are

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix},$$

$$L = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 1 \\ 1 & 0 & 0 & 1 & -2 \end{bmatrix},$$

respectively. Table I lists the considered attacks, as shown below:

The dynamic equation of the six followers is

$$\begin{cases} \dot{x}_{i,1} = 0.1cos^2(x_{i,1}) + x_{i,2}, \\ \dot{x}_{i,2} = 0.1cos^2(x_{i,1}x_{i,2}) + u_i, \\ y_i = x_{i,1}. \end{cases} \tag{69}$$

For conducting the simulation, the initial conditions of the states are chosen as $x_{i,1}(0) = 1$, $x_{i,2}(0) = 0.6$, $\hat{x}_{i,1}(0) = 1$,

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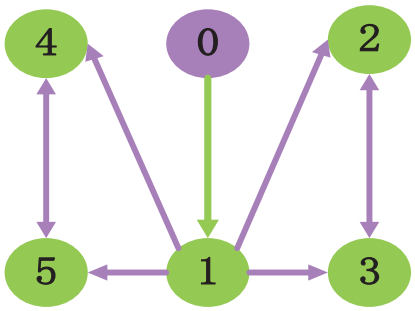


Fig. 2. Directed communication topology graph G_1.

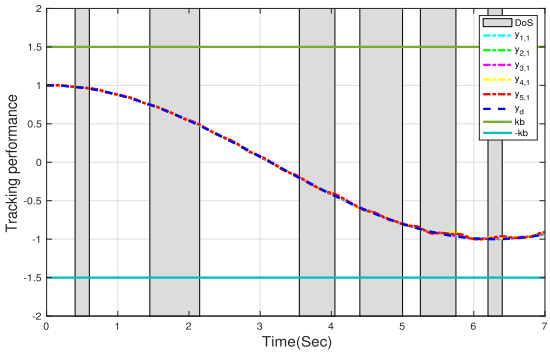


Fig. 3. The output trajectories of followers and leader.

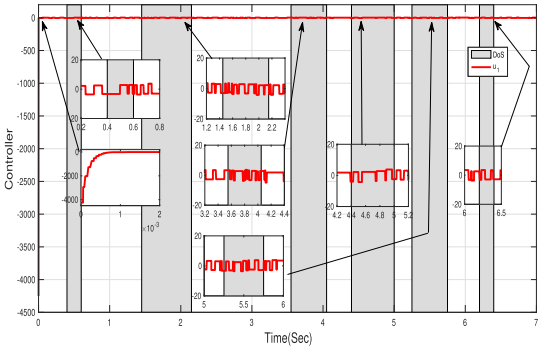


Fig. 4. The trajectories of 1th follower control signal u_1.

$\hat{x}_{i,2}(0) = 0.6$, $i = 1, \ldots, 6$. According to Assumption 3 and Assumption 4, the number and duration of the DoS attacks are chosen as $\Theta = 4$, $\tau_D = 2$, and $T = 2$. The DoS attacks frequency satisfies $n(0, 20) \leq 7$. The DoS attacks duration satisfies $D(0, 20) \leq 3.5$. The states are constrained in the regions $|x_i| < 1.5$, $i = 1, \ldots, 6$. The design parameters are selected as $C_{i,1} = 5500$, $C_{i,2} = 5500$, $l_{i,1} = 30$, $l_{i,2} = 30$, $i = 1, \ldots, 6$, $o_{1,2} = 2$,$o_{2,2} = o_{3,2} = o_{4,2} = o_{5,2} = o_{6,2} = 20$. The output of the leader is chosen as $\cos(0.5t)$.

Example 2: The MASs consisting of FDPs is considered to verify that the proposed control algorithm is right. A digraph G_2 is shown in Fig. 2 to illustrate the communication topology of the MASs, where the agent marked 0 denotes the leader and the followers are those nodes with labels ranging from 1 to 3. In the directed graph G_2, the adjacency matrix A and the

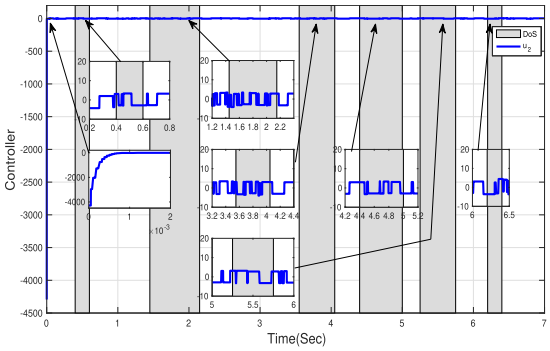


Fig. 5. The trajectories of 2th follower control signal u_2.

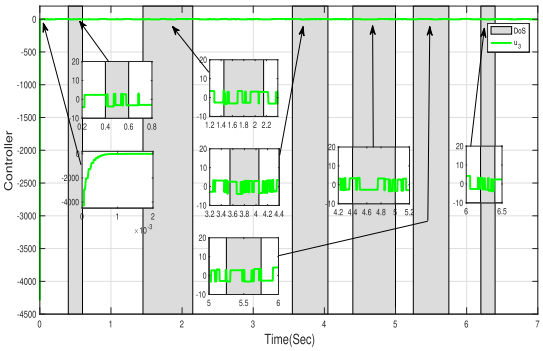


Fig. 6. The trajectories of 3th follower control signal u_3.

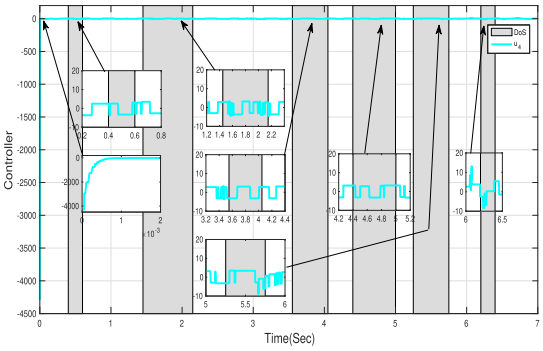


Fig. 7. The trajectories of 4th follower control signal u_4.

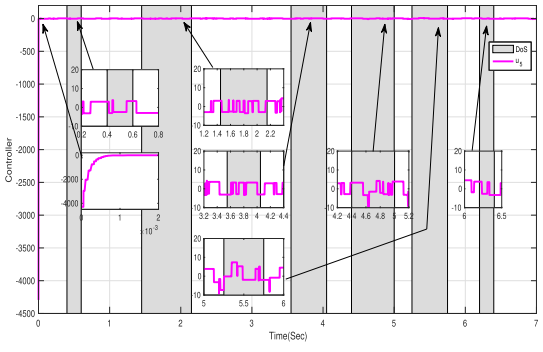


Fig. 8. The trajectories of 5th follower control signal u_5.

Laplacian matrix L are

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix},$$

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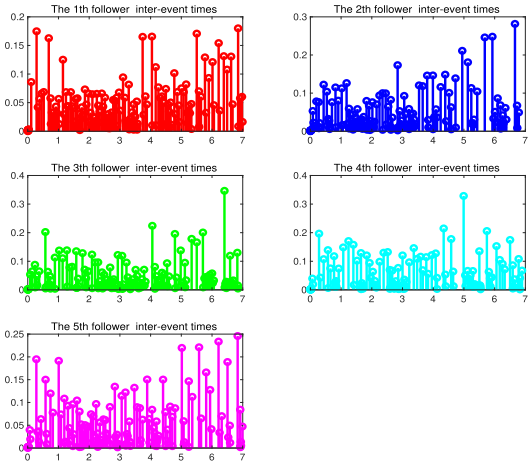


Fig. 9. The trajectories of inter-event times.

TABLE II
THE CONSIDERED DOS ATTACKS

Attack duration time (s)	Attacked edge
2.25 ∼ 3.35	(3,2)
4.85 ∼ 5.15	(2,3)
6.1 ∼ 6.5	(2,1)

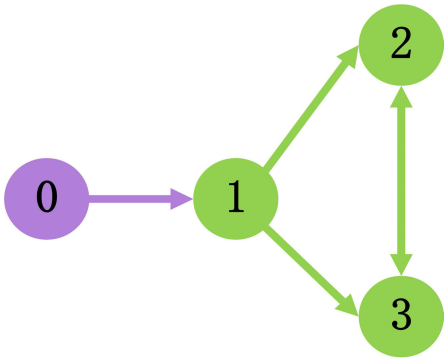


Fig. 10. Directed communication topology graph G_2.

respectively. Table II lists the considered attacks, as shown below:

The model of FDPs can is described as

$$\ddot{\eta}_i = \frac{b_i^*}{m_i l_i^2}\dot{\eta}_i - \frac{c_i^* g}{l_i}\sin(\eta_i) + u_i, \quad i = 1, 2, 3, \tag{70}$$

where l_i represents the length of pendulum, m_i represents the mass of bob, η_i represents the rod angle measured counter-clockwise from downward position, g represents the gravity acceleration, b_i^*, c_i^* denote the coefficient of damping and restoring torque, respectively. In simulations, the parameters are chosen as $g = 9.8$, $l_i = 1$, $b_i^* = -0.25$, and $c_i = 1/g$. Define $x_{i,1} = \eta_i$ and $x_{i,1} = \dot{\eta}_i$, then (70) can be rewritten as

$$\begin{cases} \dot{x}_{i,1} = x_{i,2}, \\ \dot{x}_{i,2} = -\dfrac{b_i^*}{m_i l_i^2}x_{i,2} - \dfrac{c_i^* g}{l_i}\sin(x_{i,1}) + u_i, \\ y_i = x_{i,1}, i = 1, 2, 3. \end{cases} \tag{71}$$

For conducting the simulation, the initial conditions of the states are chosen as $x_{i,1}(0) = 1$, $x_{i,2}(0) = 0.6$, $\hat{x}_{i,1}(0) = 1$, $\hat{x}_{i,2}(0) = 0.6$, $i = 1, 2, 3$. According to Assumption 3 and Assumption 4, the number and duration of the DoS attacks are chosen as $\Theta = 0.2$, $\tau_D = 5$, and $T = 4$. The DoS

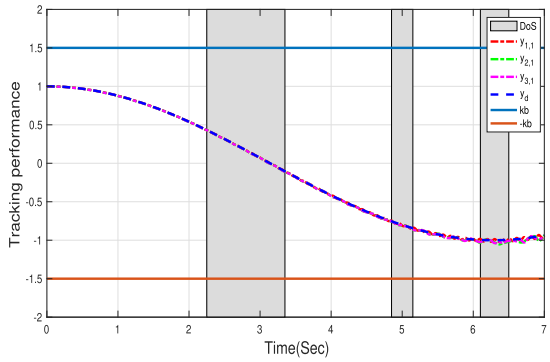


Fig. 11. The output trajectories of followers and leader.

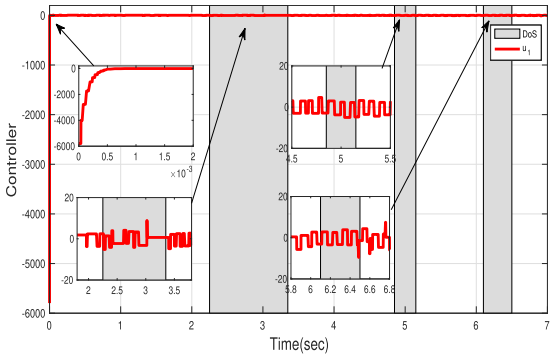


Fig. 12. The trajectories of 1th follower control signal u_1.

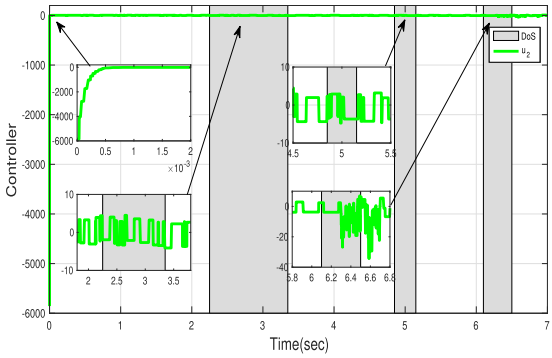


Fig. 13. The trajectories of 2th follower control signal u_2.

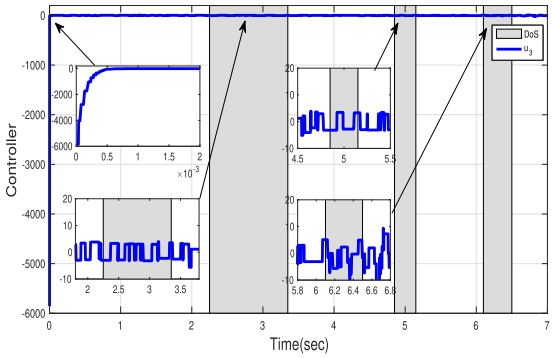


Fig. 14. The trajectories of 3th follower control signal u_3.

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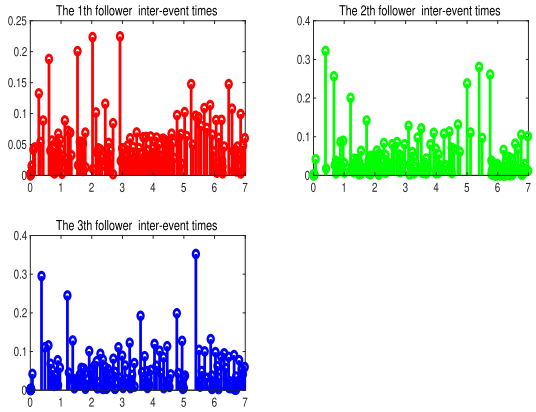


Fig. 15. The trajectories of inter-event times.

attacks frequency satisfies $n(0, 20) \leq 4$. The DoS attacks duration satisfies $D(0, 20) \leq 5$. The states are constrained in the regions $|x_i| < 1.5$, $i = 1, 2, 3$. The design parameters are selected as $C_{i,1} = 5500$, $C_{i,2} = 5500$, $l_{i,1} = l_{i,2} = 30$, $o_{1,2} = 2$, $o_{2,2} = o_{3,2} = 20$, $i = 1, 2, 3$. The output of the leader is chosen as $\cos(0.5t)$.

The simulation results are presented in Figs. 3-15. Fig. 3 and Fig. 11 show the trajectories of the system output of each follower y_i and the leader's signal y_d for the proposed control strategy in this paper. The trajectories of control signal and inter-event times of each follower are shown in Figs. 4-9 and Figs. 12-15, respectively.

VI. Conclusion

In this paper, we have considered the adaptive output feedback consensus asymptotic tracking control problem for the researched full-state-constrained nonlinear MASs under the DoS attacks and the actuator faults. To ensure the consensus performance of the attacked system, a novel distributed fuzzy secure consensus tracking control algorithm has been proposed by employing an attack detection mechanism and incorprating skillfully the Nussbaum functions. The event-triggered control strategy was introduced to save the communication resources and a state estimator was designed for each agent to estimate the immeasurable states. Moreover, to ensure the full-state constraints were not violated, the BLFs were applied in the backstepping process. In the future, we will focus on the secure bipartite consensus tracking control for nonlinear MASs under more complex attack types.

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